INTRODUCTION 1

Engineering drawing practice (technical drawing) can be categorized into three sections:

1. General principles

This is concerned with technical drawing layout, scale, line types, text, sections, projection and how to show common features.

2. Dimensioning and size tolerancing

This is concerned with the principles of dimensioning and how to apply size tolerances (not geometrical tolerances) to technical drawings. It also covers machining requirements and surface texture.

3. Geometrical tolerancing

This is concerned with controlling FORM, LOCATION and ORIENTATION of a feature (axis, plane, surface, hole etc...). This is done by applying the principles of geometrical tolerancing to engineering drawings with the aid of geometrical symbols.

This book is concerned with **Geometrical tolerancing** only.

What is a Geometrical Tolerance?

A geometrical tolerance is the maximum allowable variation of form or position of a feature. This is controlled by defining the size and shape of a tolerance zone. The specified part of the feature must be within this tolerance zone.

When to use a Geometrical Tolerance

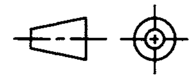
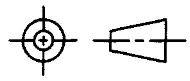
- The size tolerance of a dimension (see figure 1.3) has a certain amount of control over form and attitude (see figure 1.4), but if a better degree of control is required then geometrical tolerances should be used (see figure 1.5).

- Position of a feature is also controlled by geometrical tolerances.

- The use of geometrical tolerances can increase manufacturing costs, so they should only be used when necessary.

Orthographic Representation

Technical drawings usually consist of various two dimensional views to define an object, this is known as orthogonal projection. The two orthogonal projection methods used internationally are first angle projection and third angle projection. Third angle projection (figure 1.2) is used mainly in The United States and Canada whilst first angle projection (figure 1.1) is used mainly throughout Europe and the rest of the world. Both first and third angle projection have equal status and are approved internationally. In ISO Standard ISO1101: 2004 all figures have been drawn in first angle projection with dimensions and tolerances in millimetres. As this book is based on this standard first angle projection has been used throughout the book.

The identifying symbol for first angle projection  or for third angle projection  should be shown in the title block of your drawing.

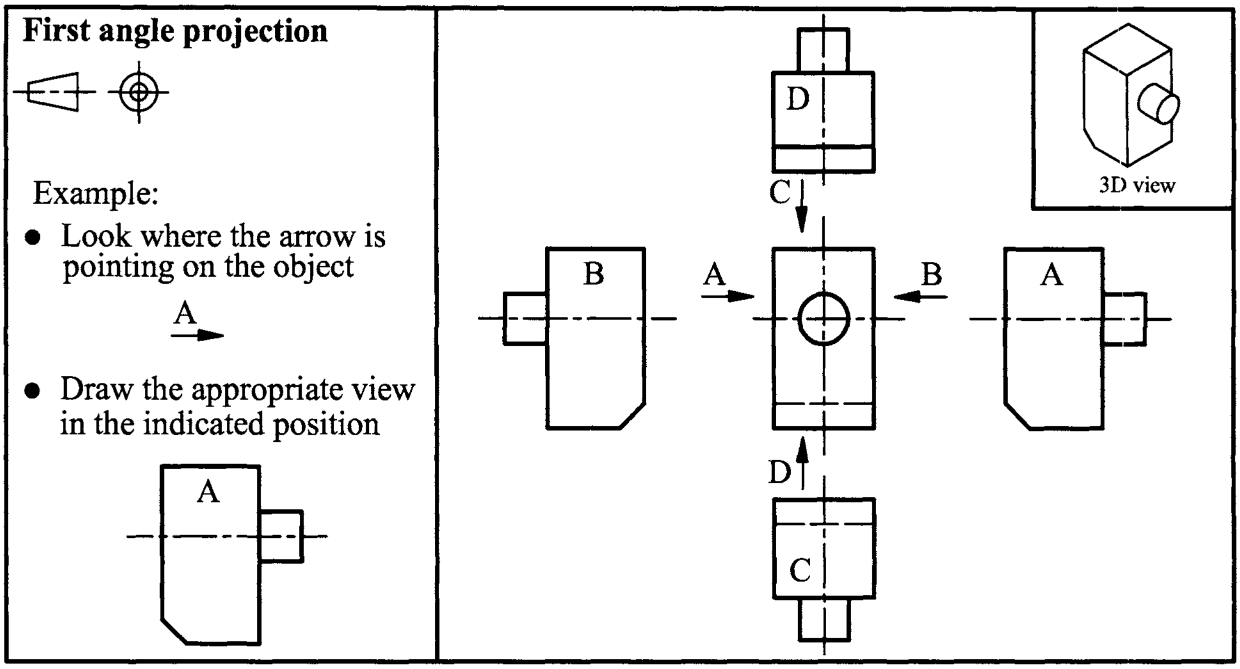


Figure 1.1

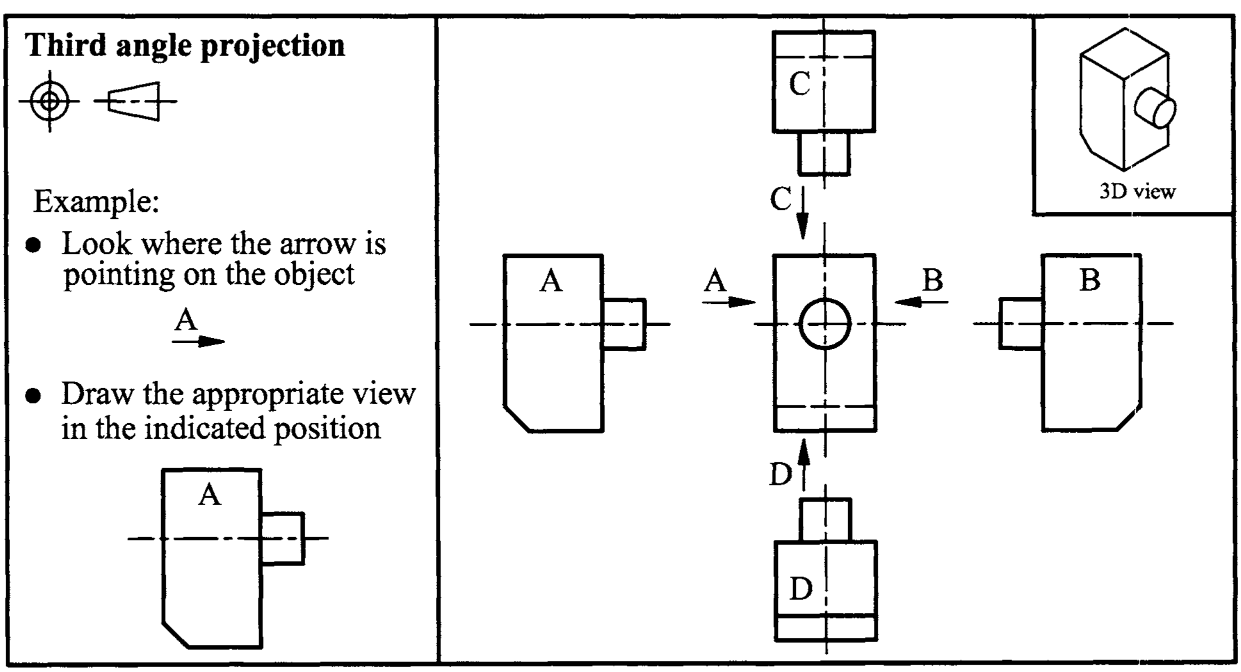
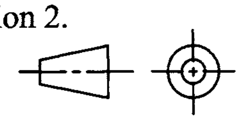


Figure 1.2

Important

- In order to use this book it is essential to read section 2.
- First angle projection is used throughout the book 

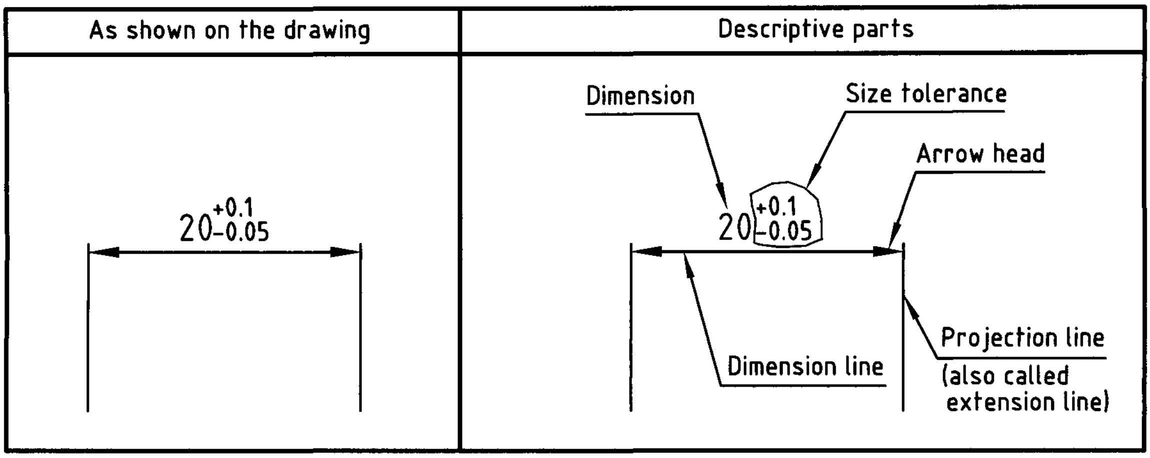


Figure 1.3 Dimension

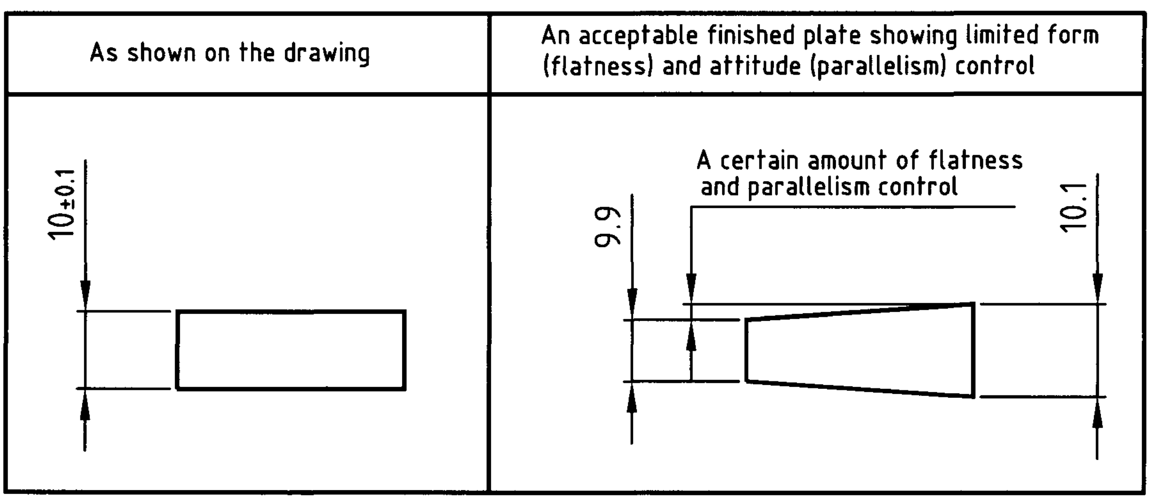


Figure 1.4 Limited form and attitude control

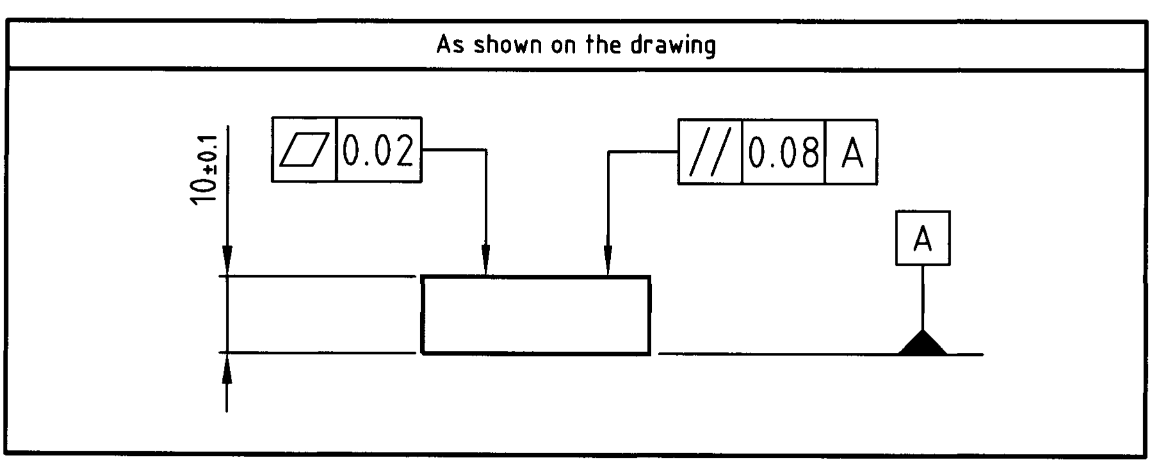


Figure 1.5 Example of geometrical tolerancing